

E-Commerce Sales Performance & Order Fulfillment Analysis

Project Overview

This project involved cleaning, analyzing, and visualizing e-commerce transaction data to evaluate sales performance, customer purchasing behavior, order fulfillment efficiency, payment preferences, and the effectiveness of marketing channels. The objective was to transform raw transactional data into actionable business intelligence that supports operational improvement and revenue growth.

1. Introduction

E-commerce businesses generate large volumes of transactional data daily. However, raw data alone does not provide meaningful business insights. This project utilized Excel, SQL Server Management Studio (SSMS), and Power BI to analyze customer orders, identify revenue drivers, evaluate fulfillment performance, and uncover opportunities to improve customer satisfaction and operational efficiency.

The analysis focused on understanding sales trends, product demand, payment behavior, referral channel performance, and order fulfillment outcomes while identifying high-value transactions and potential operational bottlenecks.

2. Business Problem

Management lacked visibility into key operational and sales performance metrics needed to support strategic decision-making.

Key business questions addressed included:

- How many orders have been processed?
- Which products generate the highest sales volume and revenue?
- Which payment methods do customers prefer?
- Which marketing channels drive the highest revenue?
- What is the average value of customer orders?
- What percentage of orders are delivered, cancelled, returned, shipped, or pending?
- Are there operational issues affecting customer satisfaction and fulfillment performance?

- Which areas present the greatest opportunities for revenue growth and process improvement?
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3. Objectives

- Analyze overall order volume and transaction activity.
 - Evaluate revenue performance across products and sales channels.
 - Identify top-performing products.
 - Assess payment method preferences.
 - Evaluate referral source effectiveness.
 - Determine average order values.
 - Analyze order fulfillment performance.
 - Identify outliers and high-value transactions.
 - Generate actionable business recommendations.
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4. Tools Used

Microsoft Excel

- Data cleaning
- Data validation
- Data quality checks
- Preliminary exploration

SQL Server Management Studio (SSMS)

- Data querying
- Data aggregation
- Revenue analysis
- Customer behavior analysis
- Order fulfillment analysis

Power BI

- Interactive dashboard development
 - KPI visualization
 - Trend analysis
 - Executive reporting
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5. Key KPIs

KPI	Value
Total Orders	1,200
Delivered Orders	231
Cancelled Orders	250
Returned Orders	247
Shipped Orders	235
Pending Orders	237
Total Revenue (Delivered Orders)	\$242,600.32
Highest Order Value	\$3,456.39
Lowest Order Value	\$11.39
Highest Unit Price	\$699.92
Lowest Unit Price	\$11.39
Top Revenue Product	Laptop
Most Ordered Product	Printer
Top Revenue Referral Source	Email
Most Frequent Referral Source	Instagram
Most Used Payment Method	Online
Highest Revenue Payment Method	Online

6. SQL Techniques Used

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

These SQL functions enabled efficient exploration, filtering, aggregation, and summarization of transactional data.

7. Key Findings

Order Volume and Fulfillment Performance

A total of 1,200 orders were analyzed.

Order status distribution revealed significant operational challenges:

Order Status	Orders	Percentage
Cancelled	250	20.83%
Returned	247	20.58%
Pending	237	19.75%
Shipped	235	19.58%
Delivered	231	19.25%

Key observation:

Only 19.25% of all orders reached successful delivery status, while over 41% of orders were either cancelled or returned. This indicates substantial inefficiencies in the order fulfillment process and potential customer satisfaction concerns.

Product Performance

- Laptop generated the highest revenue among all products.
- Printer recorded the highest order volume.
- Product-level average order value analysis identified premium products generating higher customer spend per transaction.

Revenue Performance

- Delivered orders generated total revenue of \$242,600.32.
- Revenue performance varied significantly across products, referral channels, payment methods, and months.
- Certain periods demonstrated stronger customer demand and sales performance.

Payment Method Analysis

- Online payments were the most frequently used payment option.
 - Online payments generated the highest revenue contribution.
 - Customers using online payment methods demonstrated stronger purchasing activity compared to alternative payment options.
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Referral Channel Analysis

- Email marketing generated the highest revenue among fulfilled orders.
 - Instagram generated the highest order volume.
 - Email campaigns attracted higher-value customers, while Instagram drove stronger customer acquisition.
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Customer Purchasing Behavior

- Purchase quantities varied across products and payment methods.
 - Certain products exhibited consistently stronger customer demand based on average purchase quantity.
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Outlier and High-Value Order Analysis

- Multiple high-value transactions were identified through outlier detection.
 - The Top 10 highest-value orders were isolated for review.
 - These orders may represent premium purchases, bulk orders, or high-value customer segments.
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8. Business Impact

This analysis provides decision-makers with actionable insights to improve both revenue generation and operational efficiency.

The findings help the business:

- Optimize marketing spend toward high-performing referral channels.
- Prioritize high-revenue product categories.
- Improve customer payment experiences.
- Monitor exceptional transactions.

- Enhance customer acquisition strategies.
 - Improve fulfillment performance and customer satisfaction.
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9. Recommendations

Improve Order Fulfillment Operations

With only 19.25% of orders reaching delivery status, fulfillment performance should become an immediate operational priority.

Actions:

- Investigate root causes of cancellations and returns.
- Strengthen inventory management processes.
- Improve delivery partner performance monitoring.
- Enhance order tracking visibility for customers.

Reduce Return Rates

Returned orders account for 20.58% of all transactions.

Actions:

- Improve product descriptions and specifications.
- Implement stricter quality assurance checks.
- Analyze frequently returned products for recurring issues.

Reduce Cancellation Rates

Cancelled orders represent the largest order category at 20.83%.

Actions:

- Review checkout and payment processes.
- Improve stock availability forecasting.
- Automate inventory synchronization.

Expand High-Performing Marketing Channels

Email marketing generated the highest revenue contribution.

Actions:

- Increase investment in email marketing campaigns.
- Develop personalized customer retention programs.
- Create segmented promotional campaigns targeting high-value customers.

Leverage Top Revenue Products

Laptop products generated the highest revenue.

Actions:

- Increase promotional focus on premium products.
- Introduce cross-selling and upselling opportunities.
- Bundle complementary products to increase average order value.

Strengthen Digital Payment Adoption

Online payments generated the highest revenue and usage.

Actions:

- Streamline digital payment experiences.
- Introduce incentives for digital transactions.
- Expand supported payment options.

10. Conclusion

This project successfully transformed raw e-commerce transaction data into actionable business intelligence using Excel, SQL Server, and Power BI.

The analysis identified key revenue drivers, customer purchasing patterns, payment preferences, referral channel performance, and fulfillment challenges. While products such as Laptops generated substantial revenue and Email marketing delivered strong returns, the analysis also uncovered critical operational concerns, particularly the high proportions of cancelled and returned orders.

The insights and recommendations generated from this project provide a strong foundation for improving operational efficiency, customer satisfaction, revenue growth, and strategic decision-making.